BIMU4101 – Introduction to Mobile Robotics
Midterm Project

Project Objective:

Design your own mobile robot, model it using URDF/Xacro, simulate it in Gazebo, and control its motion through ROS 2. Integrate multiple sensor types, read and visualize their data, and develop a node that uses sensor data to determine the robot's behavior.

Working Environment:

Projects will be run and evaluated using ROS 2 Lyrical and Gazebo Jetty. If you use a different version, state it in your report.

Group Formation:

- You may complete the project in groups of 2–3 students.
  - Individual projects or groups of more than 3 are not accepted.
  - Groups of 3 are preferred.
  - Complete the Google Form below to register your group members.
    - https://forms.gle/Zdcu4XHFDggFLXVh9
  - If you do not submit the form by ***19 October 2026 Monday, 23:59***, *you will be assigned to a group randomly*.

Project Requirements:

Your mobile robot should meet the following requirements:

- Design an original mobile robot for a clearly defined task or scenario. Explain your choice of locomotion system, robot dimensions and sensor placement. The robot must move in the simulated environment using ROS 2 commands.
  - Describe the robot in URDF/Xacro (or SDF): a chassis, at least two driven wheels, and any casters or extra wheels it needs.
  - Choose a drive type (differential drive is the simplest; skid-steer or others are welcome) and use the matching Gazebo plugin so the robot moves when commands are published on /cmd_vel.
  - Spawn the robot in a Gazebo world that contains obstacles (an existing world or one you build).
  - Provide a single launch file that starts the world, spawns the robot and opens RViz.
- Ready-made robot models (e.g. TurtleBot) are not accepted. Robot dimensions, component arrangement and appearance must be your own.
  - Projects using ready-made models or identical robot designs will be treated as copied work.

The sensors on the designed mobile robot should meet the following requirements:

- The minimum number of different sensor types is the number of group members + 1.
  - A 2-person group must integrate at least 3 types; a 3-person group must integrate at least 4 types.
  - Multiple sensors of the same type count as one type.
- All required sensors must be integrated into the simulated robot. Their data must be readable through ROS 2 and visualized in a suitable way (e.g. RViz, an image window or a plot).

- Examples include a camera, LiDAR, IMU and a contact sensor. Joint states, ground-truth pose and odometry published by the drive plugin alone do not count as additional sensor types.
- Each student must be responsible for integrating at least one sensor, reading its data and visualizing it. Identify these responsibilities in the report. Allocate responsibility for the additional sensor explicitly; it may be shared.

You should add a behavior to your mobile robot that meets the following requirements:

- Write at least one Python/C++ ROS 2 node that subscribes to data from at least one sensor and determines the robot's behavior. Example behaviors (choose one or propose your own):
  - Obstacle avoidance: drive forward and turn away when the LiDAR or range sensor detects something closer than a threshold.
  - Wall following at a fixed distance using LiDAR.
  - Stop and back up when the bumper/contact sensor is triggered.
  - Turn toward a colored object detected by the camera.
- In the presentation, demonstrate the behavior live in Gazebo and explain the input topic(s), message types, decision rule and motion command output of your node.

Report and Project Submission:

- Upload your code to a GitHub repository and provide the repository link in the report. The repository must be public or accessible to the instructor.
  - One shared repository per group is sufficient.
    - Create the repository under one member's GitHub account and add the other members as collaborators.
    - Every member commits their own work; the commit history should reflect each member's contribution.
  - The repository should include the source files for your ROS 2 packages, robot model, simulation configuration and any other files needed to run the project. Explain the installation and execution steps in the repository README.
- You should upload a group report named student1ID_student2ID_student3ID.pdf to CANVAS.
  - Only one submission per group is required.
- The report should include:
  - Names and student numbers of all group members, and the GitHub repository link.
  - The robot's intended task, original design, locomotion system and component connections.
  - Sensor types, placement, ROS 2 topics and message types, with evidence of data reading and visualization.
  - A contribution table identifying each student's sensor integration, data reading and visualization responsibilities, including the additional sensor.
  - Descriptions of the ROS 2 packages and nodes, including the sensor-based behavior node and its decision rule.
  - Simulation results demonstrating robot motion, sensor outputs and the effect of sensor data on robot behavior.
- **Report submission deadline:** 8 November 2026 Sunday, 23:59.
- ***Groups that do not submit the report by the deadline will not be allowed to present. All members of these groups will be considered absent from the midterm examination.***

Project Presentation:

- Presentations will take place in classroom K-Z06 on 9 November 2026 Monday, 11:10am.
  - Each group has 8 minutes for presentation and demo, followed by 2 minutes of questions.
- The explanatory part of the presentation may be based on the submitted report.
  - **Each group member must participate and explain their own contribution.**
- A live demonstration of the system is required. Demonstrate robot movement, data from all required sensor types, their visualization and the sensor-based behavior node.
  - Prepare a backup demo video in case technical problems prevent the live demonstration.
- ***Students who do not present will be considered absent from the midterm examination.***

Useful Links:

These links are provided as references. You are encouraged to consult additional sources and conduct your own research.

- https://docs.ros.org/en/lyrical/Capabilities/Simulation/URDF/URDF-Main.html
- https://gazebosim.org/docs/jetty/tutorials/
- https://gazebosim.org/api/sim/10/tutorials.html
- https://control.ros.org/lyrical/doc/getting_started/getting_started.html